

Assignment 4 - Due by Monday 5/4

You are a managing director in charge of a regional branch managing an end-to-end supply chain for a well known soda brand. You manage multiple production, reselling and retailer facilities forming a supply chain network. Historical transactions in the file *historical_transactions.csv* and the *capacities.csv* describe the historical operations of the facilities. Each facility operates on a Newsvendor-like profit $\pi_i = (D_i \wedge q_i)p_i - q_i c_i$ and $D_i = \sum_{j \neq i} p_{ij} q_j + V_i$ where V_i is a random, and possibly seasonal external demand. Each facility operates at a 60% margin, meaning $(p_i - c_i)/p_i = 6/10$ or $2p_i = 5c_i$. All the consumer facing nodes sell exactly at a price $p = 10$. You must create a python script *optimize_q.py* solving the production problem for all facilities. The workflow is as follows:

- Every month, your model is used to set the production decisions for the whole month across all the facilities. That is, you set the production values q_i for all $i = 1, \dots, N$ facilities that will be used for all the month (you can consider each q_i as a daily quantity).
- The function *optimize_quantities* should return a list with all the quantities $[q_1, \dots, q_N]$ for the month. The input of the function are all the historical transactions up to that month (a list of lists, with the origin node, target node, date, orderQuantity).
- During the month the quantities are in effect, a random demand V_i is realized for each node daily (it is 0 for the nodes that are not retailers).

To ensure that the script works as intended you are given a unit test script *unit_tests.py* ensuring that the output of your model runs within the infrastructure of the company. For the same reason, you are also constrained that the script can only use packages contained in the file *requirements.txt*.

The goal is to come up with an optimization strategy that maximizes the overall profit (of all facilities).

$$\frac{1}{T} \sum_{t=1}^T \sum_{i=1}^N \sum_{\ell=1}^{d(t)} \pi_{i,\ell},$$

Where $d(t)$ is the number of days in month t and $\pi_{i,\ell}$ is the profit of facility i on day ℓ .

Task: Come up with a model that performs best according to this metric.